COVID-19 DETECTION MILESTONE 2



By: Drew Quashie, Loleyi Oluwatomisin, Amanda Ogbonna, and Richard Alonso Garcia

MILESTONE 2 GOALS

From Milestone 1's baseline to a trained, evaluated, servable system:

- Train & fine-tune CNN models + transfer-learning architectures (85%)
- Train & fine-tune CNN models + transfer-learning architectures (85%)
- FastAPI backend endpoints (90%)
- React prediction/risk UI (70%)

THE 3-CLASS MODEL COLLAPSED

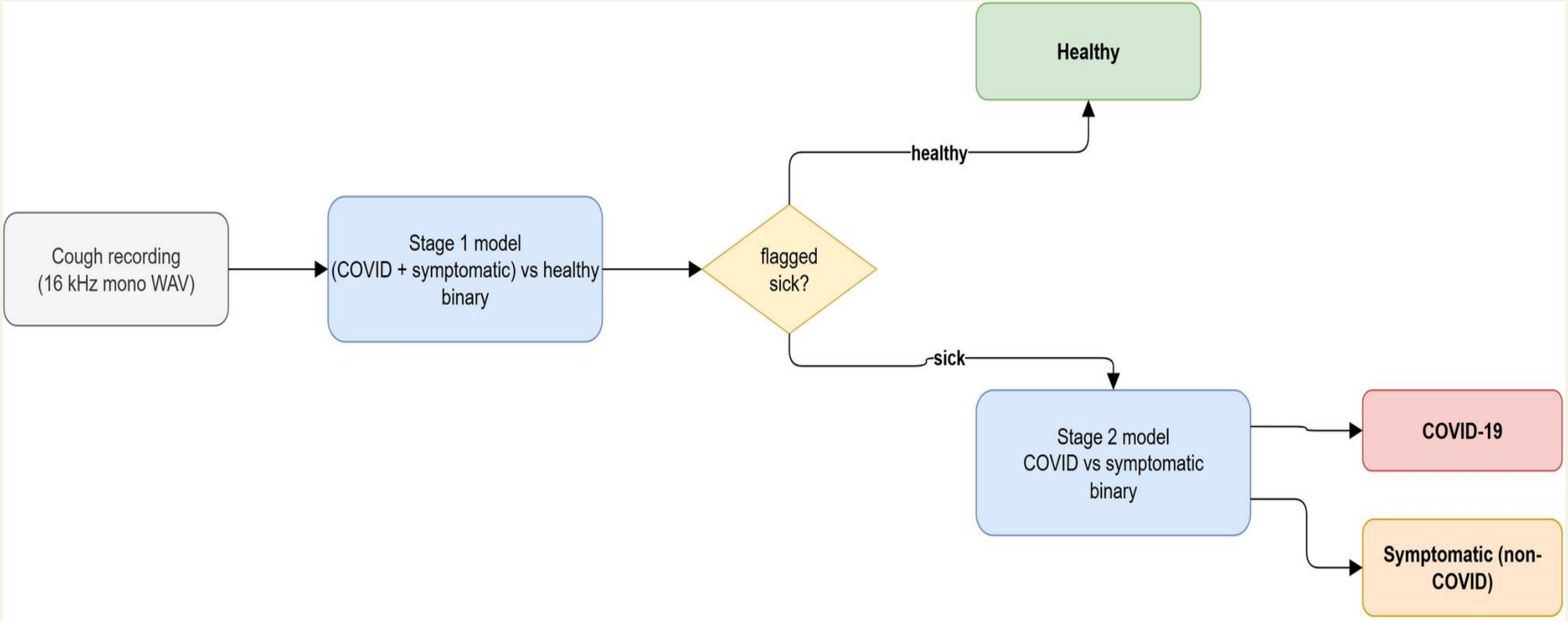
Problem:

- one 3-class CNN (healthy / symptomatic / COVID) on MFCC + log-Mel features, three input variants (wave-only, spec-only, both).

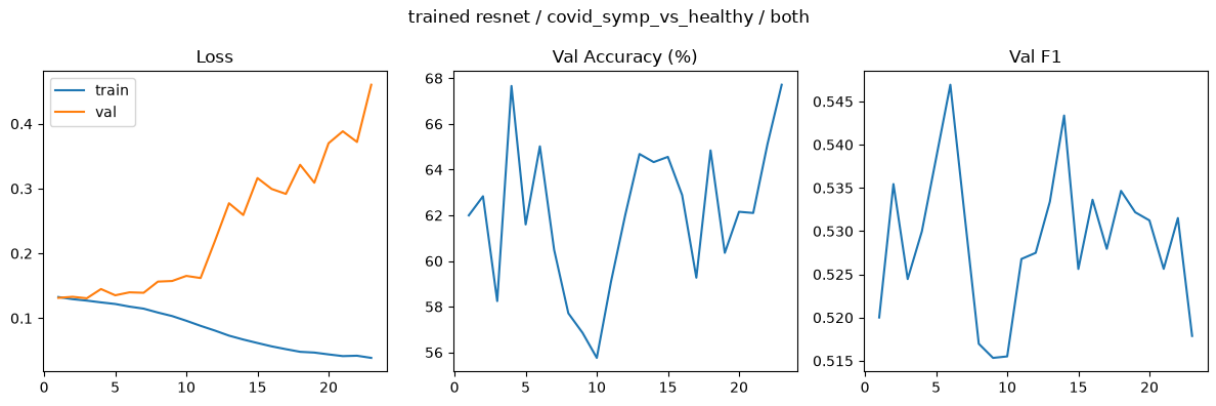
Every variant collapsed to the majority class:

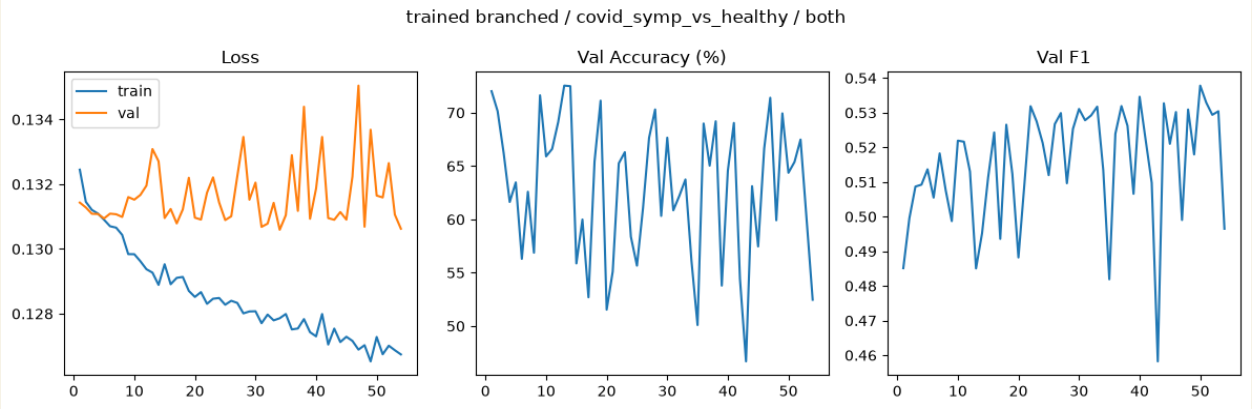
- Validation F1 frozen at **0.2459** for all 60 epochs
- Validation Accuracy stuck at **58.43%** while training loss fell and validation loss did not → memorizing, not learning

CASCADE OF MODELS SOLUTION

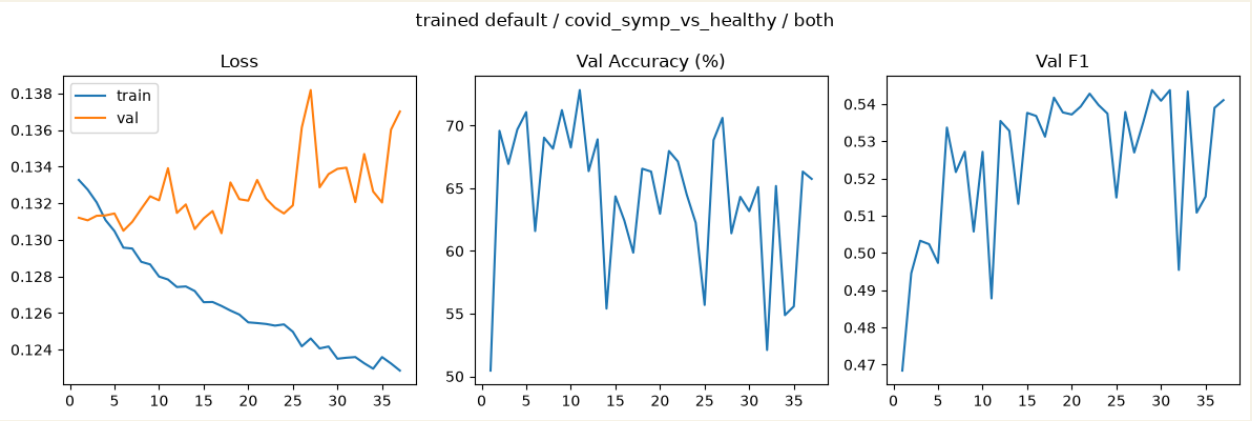


STAGE 1 RESULTS

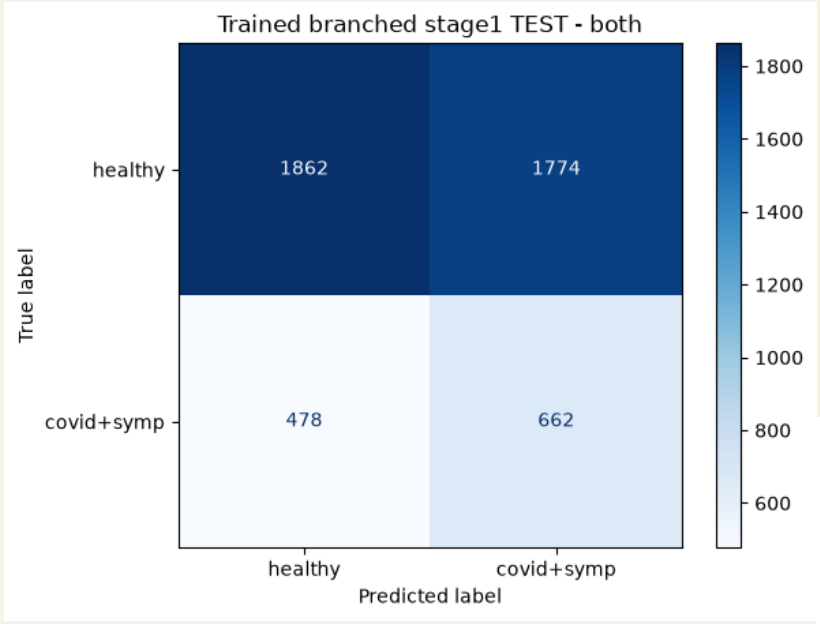


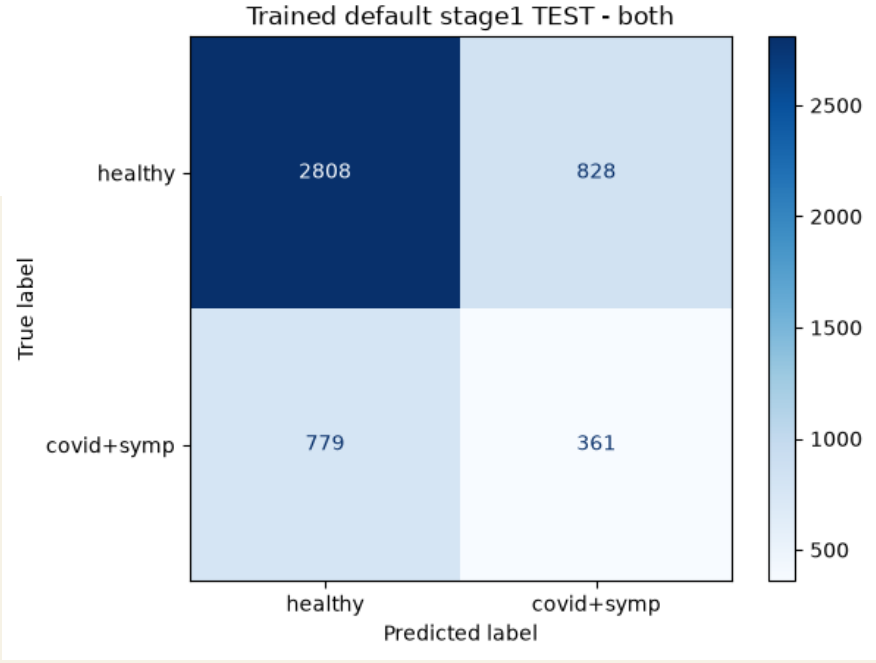


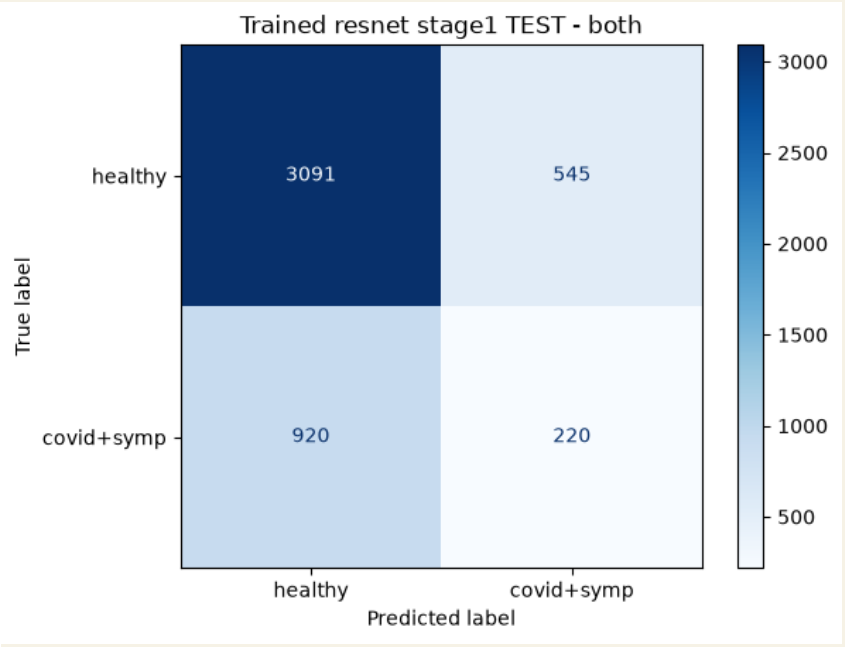
Model	Val AUC	Epochs
Default	0.57	37 (early stop)
Branched	0.56	54
ResNet50	—	23 (overfit, early-stopped)



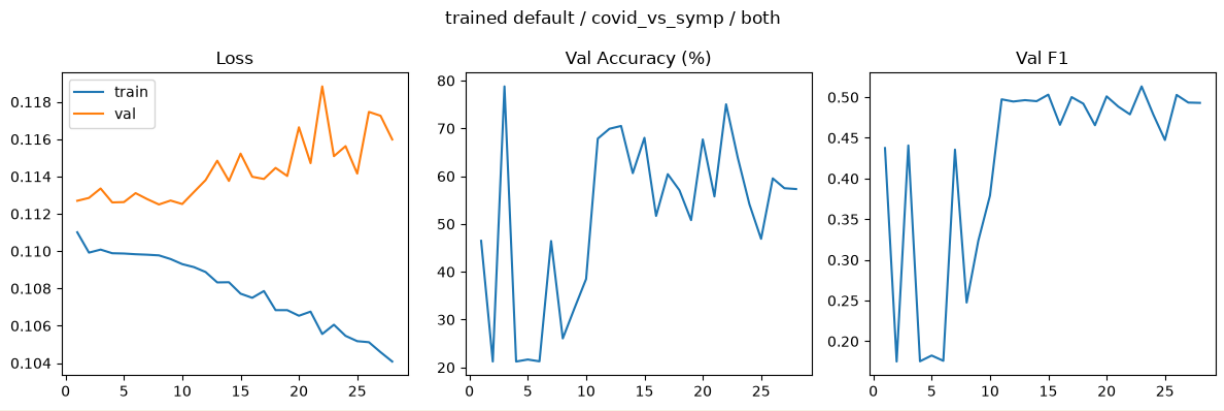
STAGE 1 CONFUSION MATRICES

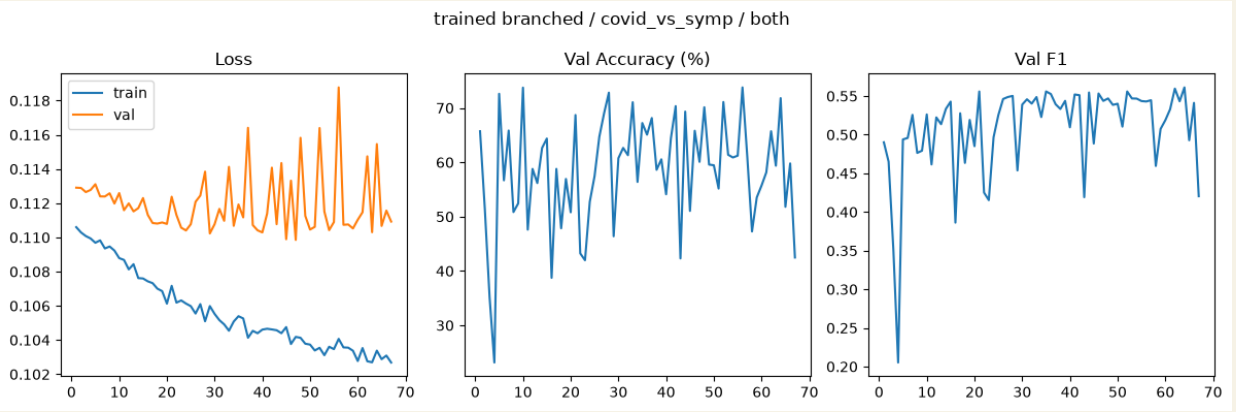




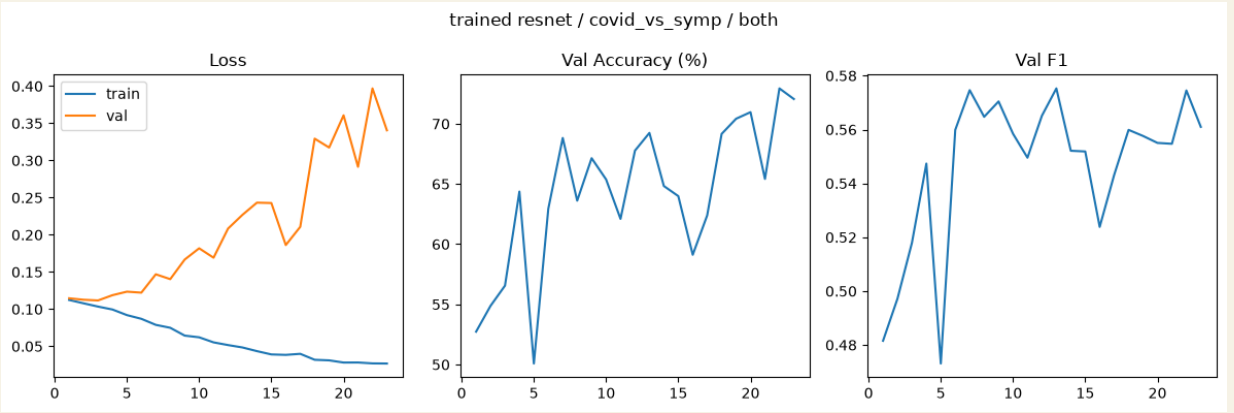


STAGE 2 RESULTS

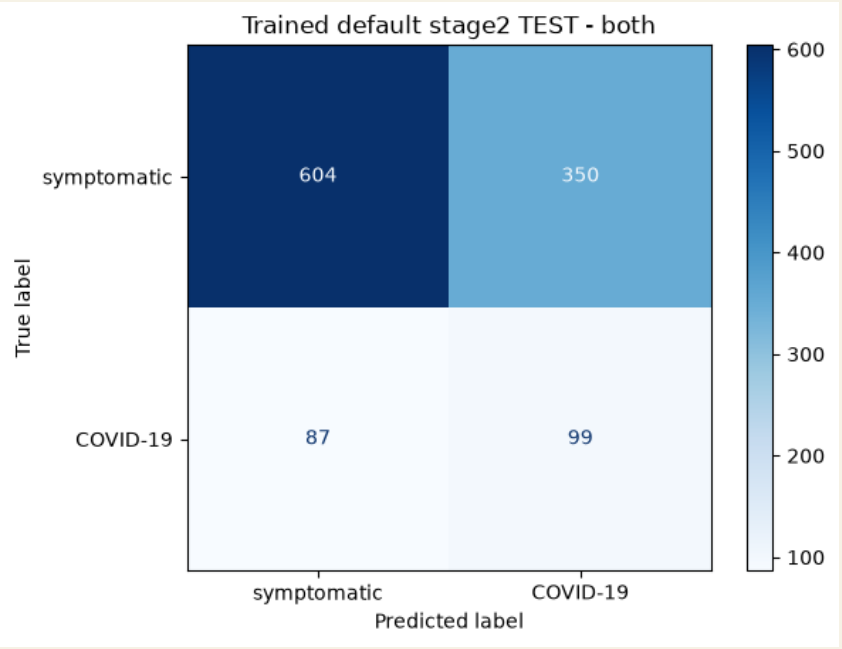


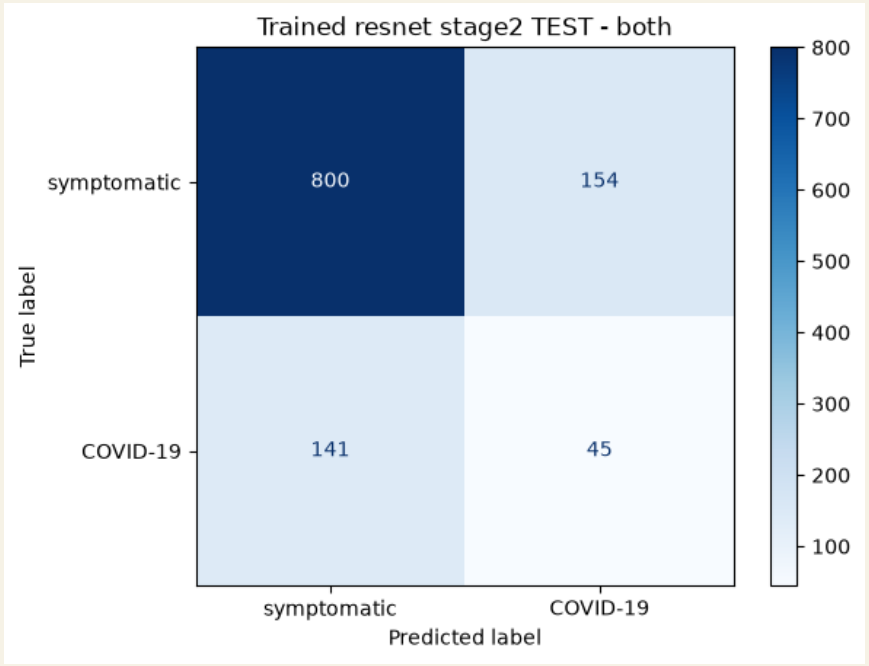


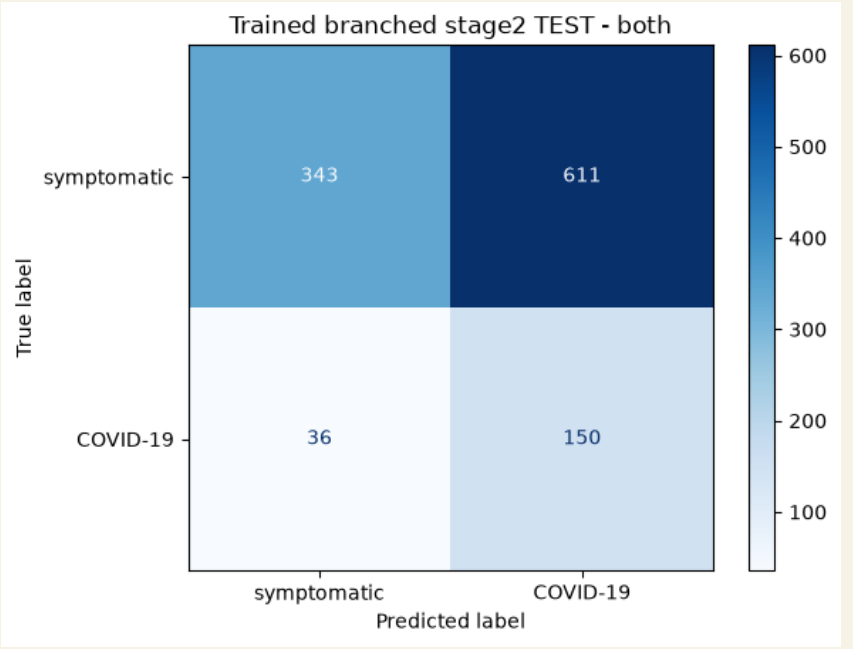
Model	Val AUC	Test AUC	COVID Recall
Default	0.532	0.599	0.47
Branched	0.623	0.637	0.61
ResNet50	0.627	0.584	0.5



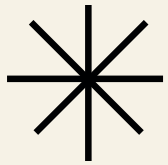
STAGE 2 CONFUSION MATRICES



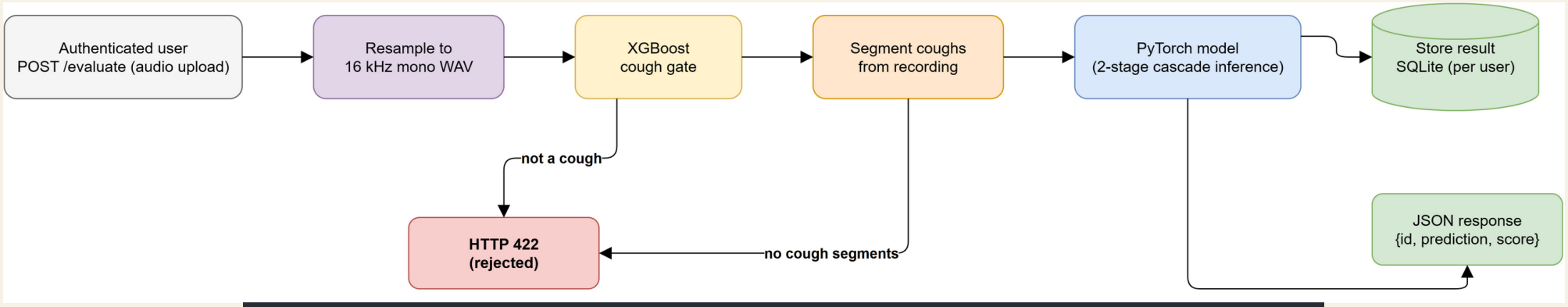




BACKEND & FRONTEND DEVELOPMENT.



FASTAPI BACKEND




```python
@router.post("/evaluate")
def evaluate(file: UploadFile = File(...), user=Depends(get_current_user)):
    raw = file.file.read()
    if not raw:
        raise HTTPException(400, "Empty upload")
    wav = to_wav_16k(raw)

    xgb_result = evaluate_xgboost(wav)
    if not is_success(xgb_result):
        raise HTTPException(422, "XGBoost gate rejected the sample")

    segments = segment_audio(wav)
    if not len(segments):
        raise HTTPException(422, "XGBoost gate rejected the sample")

    result = run_torch_model(segments)

    metadata = json.dumps({"filename": file.filename, "xgb_score": xgb_result["score"]})
    conn = get_conn()
    cur = conn.execute(
        "INSERT INTO evaluations (user_id, prediction, score, xgb_gate_success, metadata) VALUES (?, ?, ?, ?, ?)",
        (user["id"], result["prediction"], result["score"], 1, metadata),
    )
    conn.commit()
    return {"id": cur.lastrowid, "prediction": result["prediction"], "score": result["score"]}
```

DEMO



CHALLENGES

- Three-class model collapsed to majority-class predictions.
- Dataset class imbalance.
- Overfitting during longer training.
- Difficulty distinguishing similar cough patterns.
- Low COVID recall or too many false positives in some models.

WHAT'S NEXT

- Evaluate and integrate additional data sources to grow the training set
- Improve preprocessing and feature extraction for the cascade models
- Rebalance the training data and revisit the class-weighting strategy
- Refine stage 1 of the cascade: thresholds, architecture, and sick-class recall
- Retrain and re-evaluate the full cascade at the improved operating point

TEAM CONTRIBUTION & CONCLUSIONS